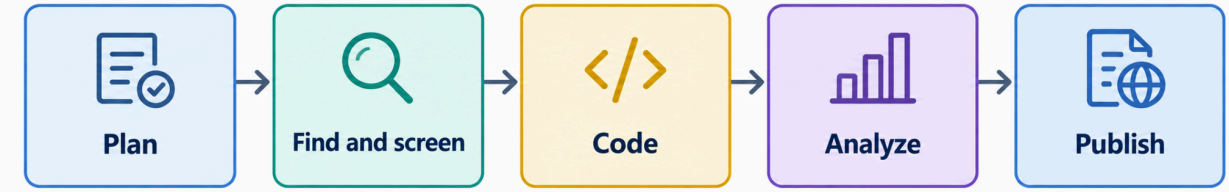
# GenAI Evidence Hub Workflow

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The GenAI Evidence Hub is a structured literature review of recent research into using Generative AI in automated scoring, item generation, and formative feedback. For more information about the study, see the pre-registration at https://osf.io/ar3yg/overview.

This page summarizes how papers move from search to the published dataset and links to the documents that define each step.

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## 1. Plan

The hypotheses, design, and analysis plan are preregistered on OSF, updated Feb 11, 2026 to remove multimodal inferences as a domain. Eligibility rules are in the Paper Selection Criteria.

## 2. Find and screen papers

Candidate papers came from existing curated repositories, researchers' private collections, hand searches of selected journals and conference proceedings, and keyword searches of

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Open Review, Google Scholar, and Springer Link: 3,390 records in all. Screening titles and abstracts against the selection criteria, then removing duplicates, pre-2023 papers, and multimodal papers, left 531. Searching was sized to the preregistered target of at least 250 coded papers rather than exhaustive coverage. Papers are managed in the Zotero library.

## 3. Code papers

Each paper is coded by a pair of coders, in rounds of about eight papers per pair:

1. Assign and verify. Papers are assigned to a pair and checked against the selection criteria at full text.
2. Pre-read. Each coder independently decides which research questions and metrics to code.
3. Alignment meeting. The pair agrees which research questions become rows and which metrics are coded.
4. Independent coding. Each coder completes their own copy of the domain's coding template.
5. Round close. Sheets are combined, inter-rater reliability (Krippendorff's alpha) is computed for each pair, and the team decides whether the coding sheet needs to be modified.
6. Consensus coding. Each pair resolves every difference from their independent codings into one row per research question on the final coding sheet, using the final version of that domain's template. Aaron Wong reviews for inconsistencies and formatting and makes the final call when a pair cannot agree.

IRR is used only to decide whether the coding sheet needs updating; the final data comes from consensus coding. Before pair coding began, the whole team coded a calibration set of papers together to refine the coding sheet; some of these were later recoded by pairs.

Coding rules are in the Training Manual and Coding Sheet Instructions. Templates: automated scoring, item generation, formative feedback. Template changes are recorded in the Coding Sheet Changelogs.

## 4. Analyze

Before analysis, the final coding sheet is deduplicated to one entry per study and model names are standardized. The analysis follows the preregistration: descriptive statistics on which models and metrics are used (hypotheses 1 and 2), and a Bayesian hierarchical Bradley-Terry model of within-study model rankings (hypothesis 3).

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Part of the GenAI Evidence Hub, a Learning Data Insights project reviewing research on evaluation practices for generative AI in educational assessment. © 2026 Learning Data Insights. Licensed under CC BY 4.0.

## 5. Publish

The annotated dataset, codebook, and synthesis report are released under CC BY 4.0 on the GenAI Insights Hub. Coded papers will also be browsable in the AI Evidence Hub library, filtered by domain and coding status.

## Use of AI in this review

Humans conducted all review and coding; AI was not used for the initial process. Once human ground truth was established, the project explored where AI could help, and released what worked as public goods:

- triageQ screens papers against a criteria profile. It was built during these experiments, was not used for inclusion decisions, and is released as a prototype and reference implementation.
- IRR calculator is a standalone app for calculating Krippendorff's alpha, released on GitHub so reported reliability values can be reproduced.
- LLM disagreement diagnostics were piloted in early 2026 and not adopted, because the models tested could not reliably compute alpha.
- Insights extraction was explored and dropped: coders themselves disagree about which findings matter, so there is no reliable standard for an LLM to match.

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